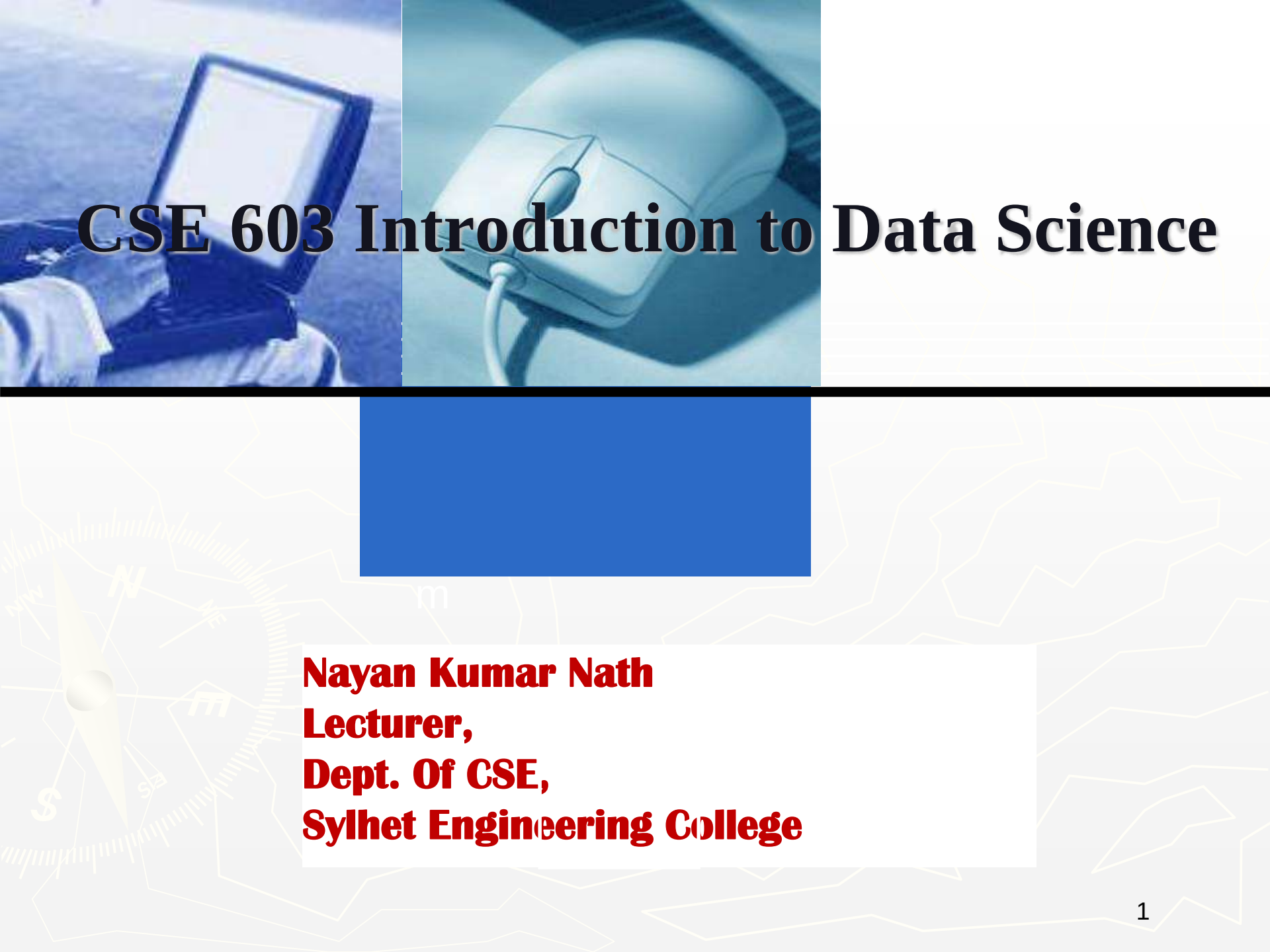




CSE 603 Introduction to Data Science

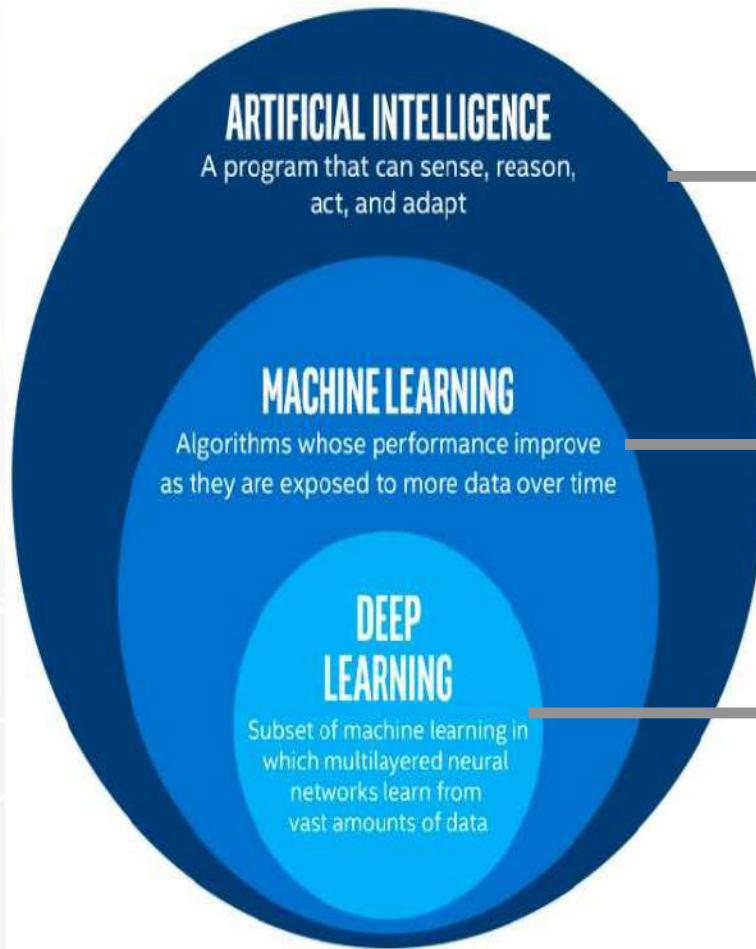
Nayan Kumar Nath
Lecturer,
Dept. Of CSE,
Sylhet Engineering College

What is machine learning?



Blue faces seem to be important...

What is machine learning really?



1950's: creation of first “intelligent” algorithms and programs. Alan Turing published “Computer Machinery and Intelligence” which proposed a test of machine intelligence called The Imitation Game

1980's: statistical models and algorithms that can learn from data

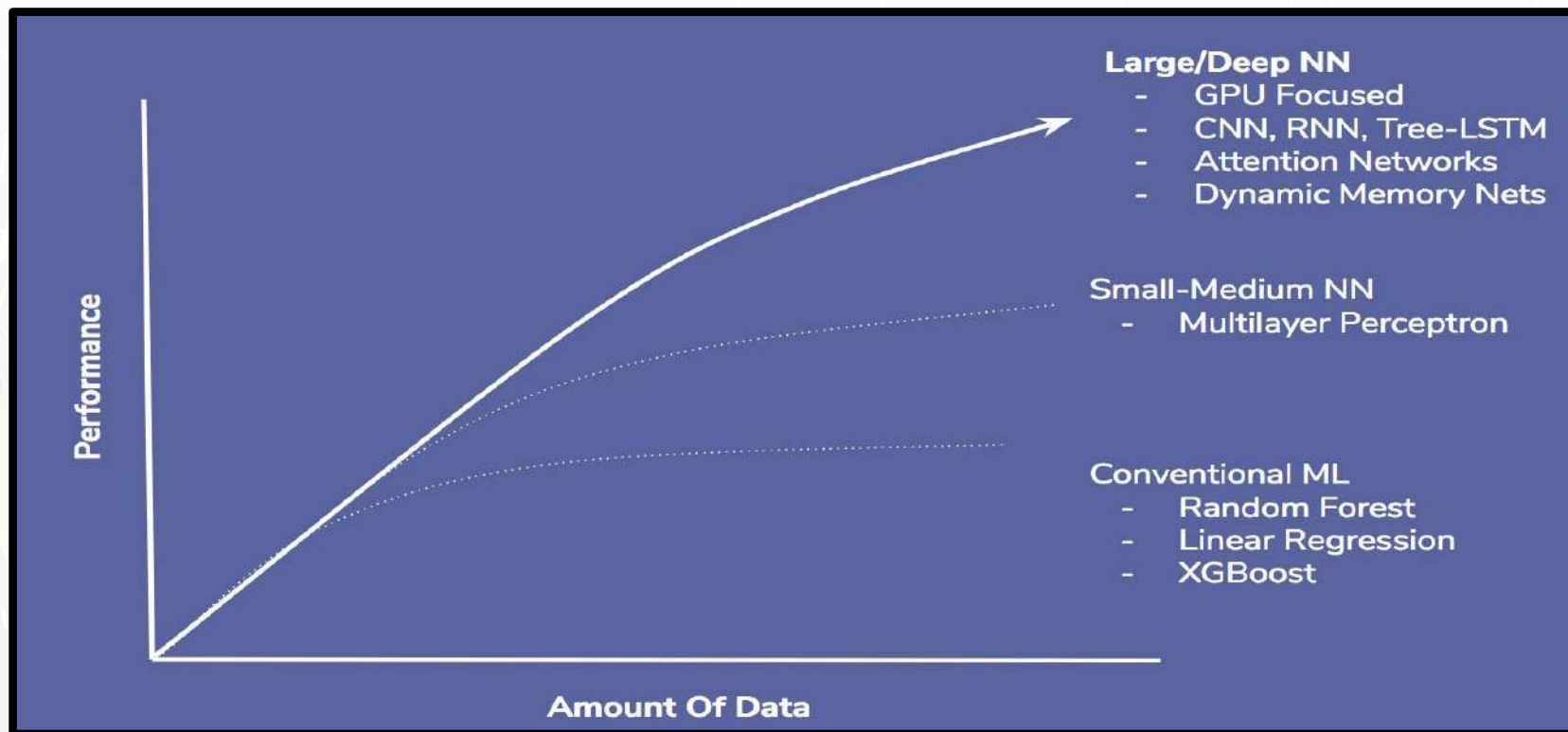
2010's: statistical models and algorithms inspired by neurones that can learn from data

Part II: Deep Learning

Deep learning (DL) is the type of machine learning (ML) that resembles human brains, where it learns from data by using artificial neural networks. Just like human brains, these deep neural networks learn from real-life examples. In a few cases, it has surpassed human intelligence, just like Google's AlphaGo has defeated number one Go Player Ke Jie.

Why now? In recent years two things became available:

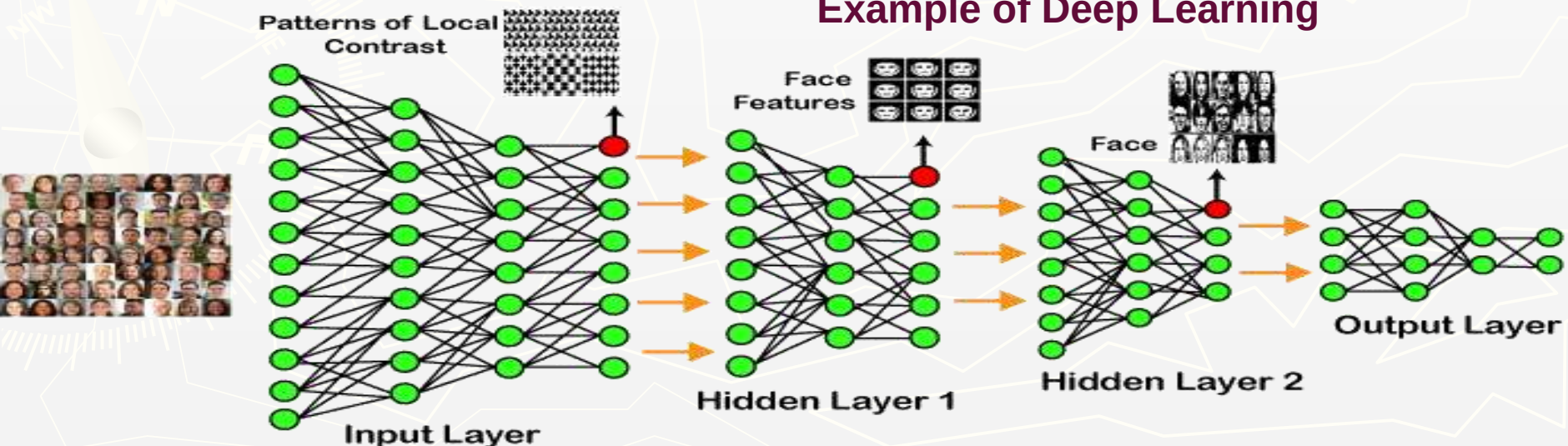
1. A lot of data



What is deep learning?

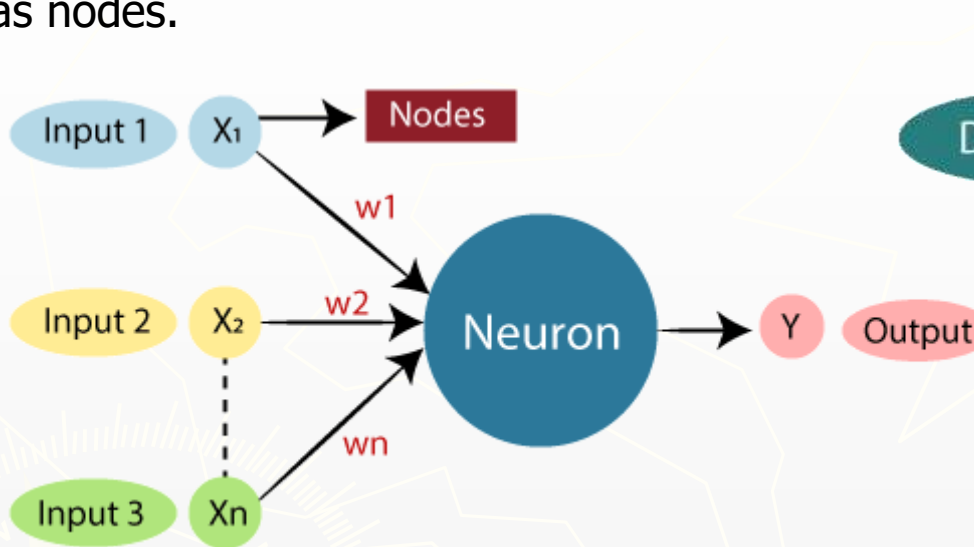
Deep Learning is a part of Machine Learning that uses artificial neural networks to learn from lots of data without needing explicit programming. These networks are inspired by the human brain and can be used for things like recognizing images, understanding speech, and processing language. **Imagine teaching a computer to recognize cats: instead of telling it to look for whiskers, ears, and a tail, you show it thousands of pictures of cats. The computer finds the common patterns all by itself and learns how to identify a cat. This is the essence of deep learning.** There are different types of deep learning networks, like feedforward neural networks, convolutional neural networks, and recurrent neural networks. Deep Learning needs lots of labeled data and powerful computers to work well, but it can achieve very good results in many applications. Deep learning is a collection of statistical techniques of machine learning for learning feature hierarchies that are actually based on artificial neural networks. So basically, deep learning is implemented by the help of deep networks, which are nothing but neural networks with multiple hidden layers.

Example of Deep Learning

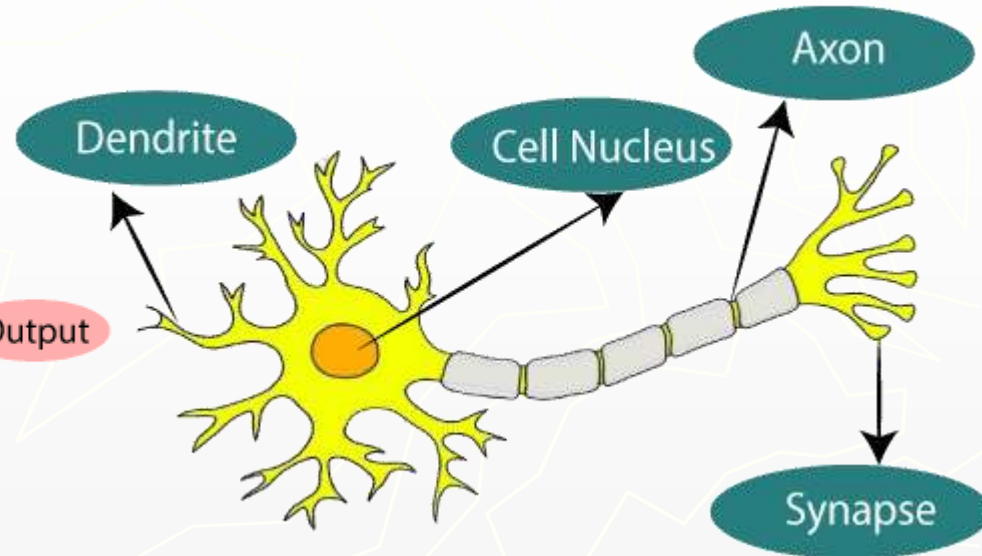


What is Artificial Neural Network?

The term "**Artificial Neural Network**" is derived from Biological neural networks that develop the structure of a human brain. Similar to the human brain that has neurons interconnected to one another, artificial neural networks also have neurons that are interconnected to one another in various layers of the networks. These neurons are known as nodes.



Artificial Neural Network.



Biological Neural Network.

Biological Neural Network	Artificial Neural Network
Dendrites	Inputs
Cell nucleus	Nodes
Synapse	Weights
Axon	Output

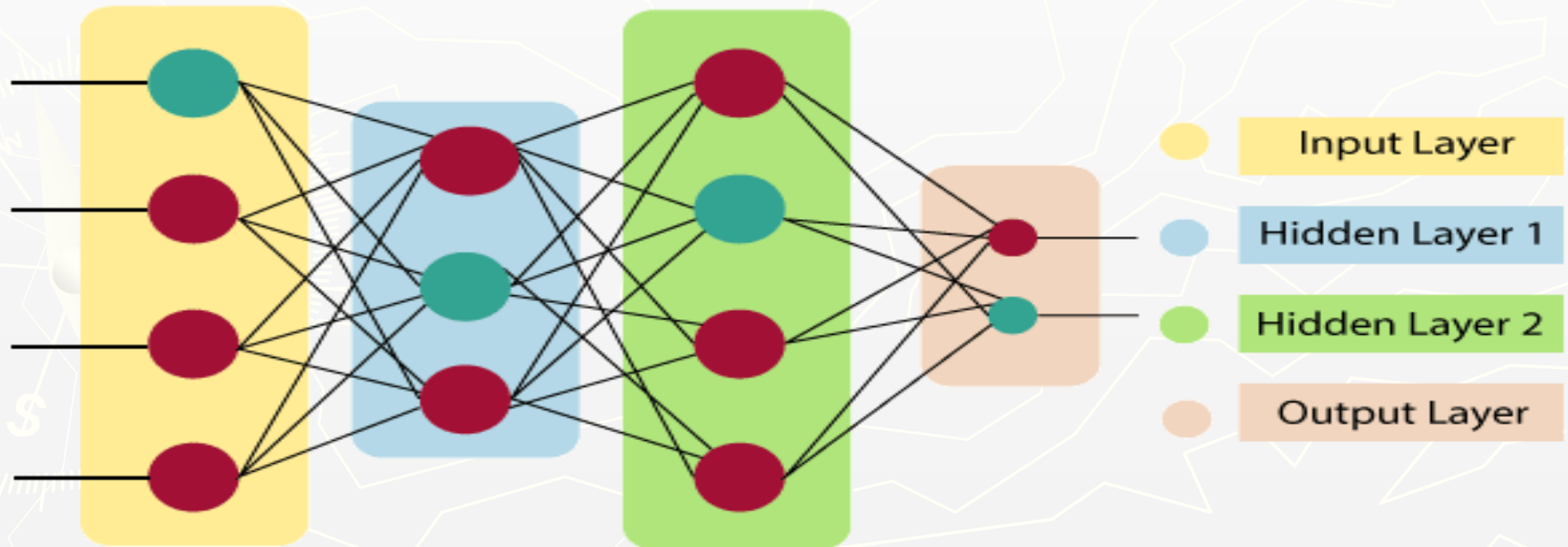
Key Components of an Artificial Neural Network:

- **Neurons (Nodes):**
 - The basic building blocks of an ANN.
 - Each neuron receives input, processes it, and produces an output.
 - Neurons are organized into layers.
- **Layers:**
 - **Input Layer:** Receives the initial data (features) and passes it to the next layer.
 - **Hidden Layers:** Intermediate layers between the input and output layers. They perform computations and transformations on the data.
 - **Output Layer:** Produces the final result (e.g., a classification or prediction).
- **Weights and Biases:**
 - **Weights:** Parameters that determine the strength of the connection between neurons.
 - **Biases:** Additional parameters that allow the model to shift the output along the activation function.
- **Activation Function:**
 - Introduces non-linearity into the network, enabling it to learn complex patterns.
 - Common activation functions include:
 - ReLU (Rectified Linear Unit)
 - Sigmoid
 - Tanh (Hyperbolic Tangent)
 - Softmax (for output layers in classification tasks).
- **Loss Function:**
 - Measures the difference between the predicted output and the actual target.
 - Common loss functions include:
 - Mean Squared Error (MSE) for regression.
 - Cross-Entropy Loss for classification.
- **Optimization Algorithm:**
 - Adjusts the weights and biases to minimize the loss function.
 - Common optimization algorithms include:
 - Gradient Descent
 - Stochastic Gradient Descent (SGD)
 - Adam, RMSprop, etc.
- **Backpropagation:**
 - A method for training the network by propagating errors backward from the output layer to the input layer.
 - It updates the weights and biases using gradient descent.

The architecture of an artificial neural network:

The artificial neural network is designed by programming computers to behave simply like interconnected brain cells. There are around 1000 billion neurons in the human brain. Each neuron has an association point somewhere in the range of 1,000 and 100,000. In the human brain, data is stored in such a manner as to be distributed, and we can extract more than one piece of this data when necessary from our memory parallelly. We can say that the human brain is made up of incredibly amazing parallel processors.

To understand the concept of the architecture of an artificial neural network, we have to understand what a neural network consists of. In order to define a neural network that consists of a large number of artificial neurons, which are termed units arranged in a sequence of layers. Lets us look at various types of layers available in an artificial neural network. Artificial Neural Network primarily consists of three layers:



The architecture of an artificial neural network:

- **Input Layer:**

As the name suggests, it accepts inputs in several different formats provided by the programmer.

- **Hidden Layer:**

The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.

- **Output Layer:**

The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer.

The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.

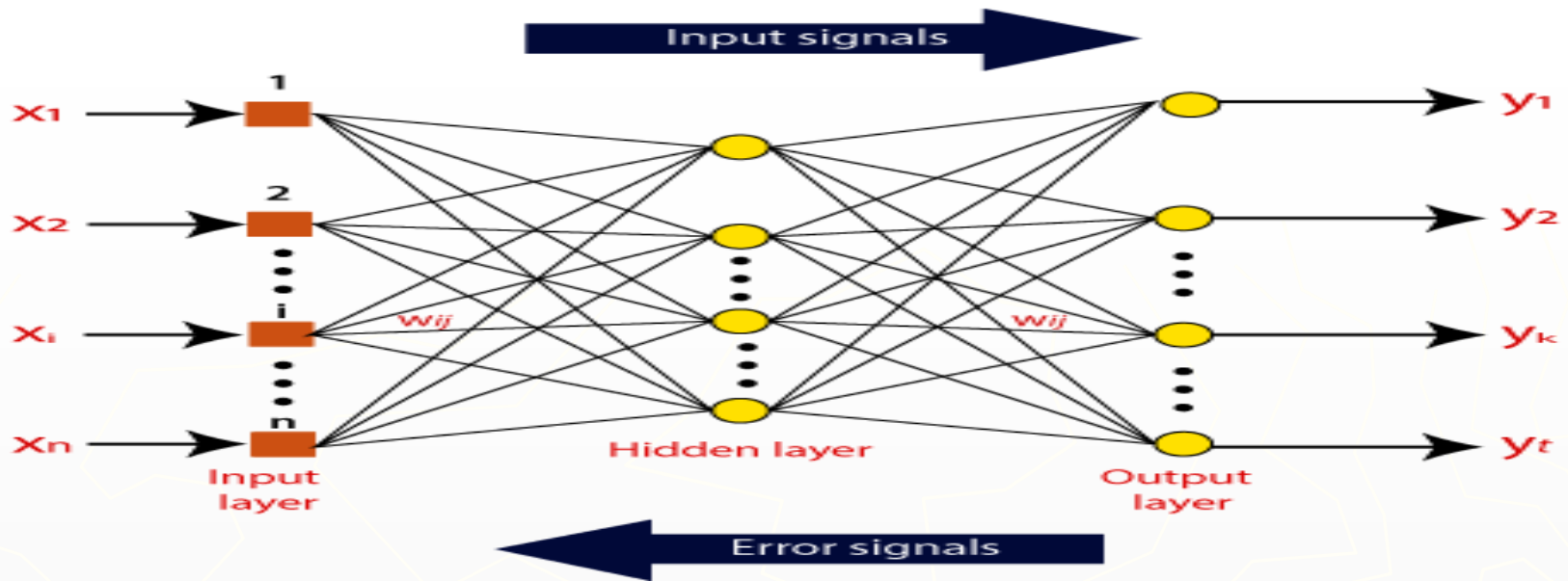
$$\sum_{i=1}^n W_i * X_i + b$$

It determines weighted total is passed as an input to an activation function to produce the output. Activation functions choose whether a node should fire or not. Only those who are fired make it to the output layer. There are distinctive activation functions available that can be applied upon the sort of task we are performing.

How do artificial neural networks work?

Artificial Neural Network can be best represented as a weighted directed graph, where the artificial neurons form the nodes. The association between the neurons outputs and neuron inputs can be viewed as the directed edges with weights. The Artificial Neural Network receives the input signal from the external source in the form of a pattern and image in the form of a vector. These inputs are then mathematically assigned by the notations $x(n)$ for every n number of inputs.

The architecture of an artificial neural network:



- Afterward, each of the input is multiplied by its corresponding weights (these weights are the details utilized by the artificial neural networks to solve a specific problem). In general terms, these weights normally represent the strength of the interconnection between neurons inside the artificial neural network. All the weighted inputs are summarized inside the computing unit.
- If the weighted sum is equal to zero, then bias is added to make the output non-zero or something else to scale up to the system's response. Bias has the same input, and weight equals to 1. Here the total of weighted inputs can be in the range of 0 to positive infinity. Here, to keep the response in the limits of the desired value, a certain maximum value is benchmarked, and the total of weighted inputs is passed through the activation function(**An activation function is a mathematical function applied to the output of a neuron. It introduces non-linearity into the model, allowing the network to learn and represent complex patterns in the data. Without this non-linearity feature, a neural network would behave like a linear regression model, no matter how many layers it has.**). The activation function refers to the set of transfer functions used to achieve the desired output.

The architecture of an artificial neural network:

There is a different kind of the activation function, but primarily either linear or non-linear sets of functions. Some of the commonly used sets of activation functions are the Binary, linear, and Tan hyperbolic sigmoidal activation functions. Let us take a look at each of them in details:

Binary: In binary activation function, the output is either a one or a 0. Here, to accomplish this, there is a threshold value set up. If the net weighted input of neurons is more than 1, then the final output of the activation function is returned as one or else the output is returned as 0.

Sigmoidal Hyperbolic: The Sigmoidal Hyperbola function is generally seen as an "S" shaped curve. Here the tan hyperbolic function is used to approximate output from the actual net input. The function is defined as:

$$F(x) = (1 / (1 + \exp(-\alpha x)))$$

Where α is considered the Steepness parameter.

Types of Artificial Neural Network:

There are various types of Artificial Neural Networks (ANN) depending upon the human brain neuron and network functions, an artificial neural network similarly performs tasks. The majority of the artificial neural networks will have some similarities with a more complex biological partner and are very effective at their expected tasks. For example, segmentation or classification.

Feedback ANN(recurrent neural network): A feedback neural network, often referred to as a **recurrent neural network (RNN)**, allows information to flow in cycles. The network has **memory** and can retain information about previous inputs, making it suitable for sequential data. Neurons in hidden layers receive input not only from the current input but also from their own previous outputs.

Feed-Forward ANN(): A feed-forward network is a basic neural network comprising of an input layer, an output layer, and at least one layer of a neuron. Through assessment of its output by reviewing its input, the intensity of the network can be noticed based on group behavior of the associated neurons, and the output is decided. The primary advantage of this network is that it figures out how to evaluate and recognize input patterns.

Types of Neural Networks:

- **Feedforward Neural Networks (FNN):**
 - The simplest type, where data flows in one direction from input to output.
 - No cycles or loops.
- **Convolutional Neural Networks (CNN):**
 - Specialized for processing grid-like data (e.g., images).
 - Uses convolutional layers to extract spatial features.
- **Recurrent Neural Networks (RNN):**
 - Designed for sequential data (e.g., time series, text).
 - Contains loops to allow information to persist over time.
 - Variants include LSTM (Long Short-Term Memory) and GRU (Gated Recurrent Unit).
- **Generative Adversarial Networks (GAN):**
 - Consists of two networks (generator and discriminator) that compete to generate realistic data.
- **Autoencoders:**
 - Used for unsupervised learning and dimensionality reduction.
 - Compresses input data into a latent representation and reconstructs it.

Types of Deep Learning Networks



1. Feed Forward Neural Network

A feed-forward neural network is none other than an [Artificial Neural Network](#), which ensures that the nodes do not form a cycle. In this kind of neural network, all the **perceptrons** (The perceptron is an artificial neuron that attempts to approximate a biological neuron. The perceptron takes one or more weighted inputs and returns a single binary output, either 1 or 0.) are organized within layers, such that the input layer takes the input, and the output layer generates the output. Since the hidden layers do not link with the outside world, it is named as hidden layers. Each of the **perceptrons** contained in one single layer is associated with each node in the subsequent layer. It can be concluded that all of the nodes are fully connected. It does not contain any visible or invisible connection between the nodes in the same layer. There are no back-loops in the feed-forward network. To minimize the prediction error, the backpropagation algorithm can be used to update the weight values.

Applications: Data Compression, Pattern Recognition, Computer Vision, Sonar Target Recognition, Speech Recognition, Handwritten Characters Recognition

2. Recurrent Neural Network

[Recurrent neural networks](#) are yet another variation of feed-forward networks. Here each of the neurons present in the hidden layers receives an input with a specific delay in time. The Recurrent neural network mainly accesses the preceding info of existing iterations. For example, to guess

Types of Deep Learning Networks

the succeeding word in any sentence, one must have knowledge about the words that were previously used. It not only processes the inputs but also shares the length as well as weights crossways time. It does not let the size of the model to increase with the increase in the input size. However, the only problem with this recurrent neural network is that it has slow computational speed as well as it does not contemplate any future input for the current state. It has a problem with reminiscing prior information.

Applications: Machine Translation, Robot Control, Time Series Prediction, Speech Recognition Speech Synthesis, Time Series Anomaly Detection, Rhythm Learning, Music Composition

3. Convolutional Neural Network

Convolutional Neural Networks are a special kind of neural network mainly used for image classification, clustering of images and object recognition. DNNs enable unsupervised construction of hierarchical image representations. To achieve the best accuracy, deep convolutional neural networks are preferred more than any other neural network.

Applications: Identify Faces, Street Signs, Tumors, Image Recognition, Video Analysis, NLP, Anomaly Detection, Drug Discovery, Checkers Game, Time Series Forecasting.

4. Restricted Boltzmann Machine

RBMs are yet another variant of Boltzmann Machines(A Boltzmann Machine is a network of symmetrically connected, neuron- like units that make stochastic decisions about whether to be on or off. Boltzmann machines have a simple learning algorithm that allows them to discover interesting features in datasets composed of binary vectors.). Here the neurons present in the input layer and the hidden layer encompasses symmetric connections amid them. However, there is no internal association within the respective layer. But in contrast to RBM, Boltzmann machines do encompass internal connections inside the hidden layer. These restrictions in BMs helps the model to train efficiently.

Applications: Filtering, Feature Learning, Classification, Risk Detection, Business and Economic analysis.

Deep Learning Networks

5. Autoencoders

An autoencoder neural network is another kind of unsupervised machine learning algorithm. Here the number of hidden cells is merely small than that of the input cells. But the number of input cells is equivalent to the number of output cells. An autoencoder network is trained to display the output similar to the fed input to force AEs to find common patterns and generalize the data. The autoencoders are mainly used for the smaller representation of the input. It helps in the reconstruction of the original data from compressed data. This algorithm is comparatively simple as it only necessitates the output identical to the input.

- Encoder:** Convert input data in lower dimensions.

- Decoder:** Reconstruct the compressed data.

Applications: Classification, Clustering, Feature Compression.

Deep learning applications

Self-Driving Cars

In self-driven cars, it is able to capture the images around it by processing a huge amount of data, and then it will decide which actions should be incorporated to take a left or right or should it stop. So, accordingly, it will decide what actions it should take, which will further reduce the accidents that happen every year.

Voice Controlled Assistance

When we talk about voice control assistance, then **Siri** is the one thing that comes into our mind. So, you can tell Siri whatever you want it to do it for you, and it will search it for you and display it for you.

Automatic Image Caption Generation

Whatever image that you upload, the algorithm will work in such a way that it will generate caption accordingly. If you say blue colored eye, it will display a blue-colored eye with a caption at the bottom of the image.

Automatic Machine Translation

With the help of automatic machine translation, we are able to convert one language into another with the help of deep learning.

Deep Learning Networks

Limitations

- It only learns through the observations.
- It comprises of biases issues.

Advantages

- It lessens the need for feature engineering.
- It eradicates all those costs that are needless.
- It easily identifies difficult defects.
- It results in the best-in-class performance on problems.

Disadvantages

- It requires an ample amount of data.
- It is quite expensive to train.
- It does not have strong theoretical groundwork.



What is Time-Series Analysis?

An ordered set of observations with respect to time periods is a time series. In simple words, a sequential organization of data accordingly to their **time of occurrence is termed as time series**. For example, "how do people get to know that the price of an object as increased or decreased over time", they do so **by comparing the price of an object over a set of the time period**. A **time series data** is the set of measurements taking place in a constant interval of time, here time acts as independent variable and the objective (to study changes in a characteristics) is dependent variables.

For example, one can measure - Consumption of energy per hour, Sales on daily basis, Company's profits per quarter, Annual changes in a population of a country.

The time series data is of three types:

- **Time series data**: A set of observations contains values, taken by variable at different times.
- **Cross-sectional data**: Data values of one or more variables, gathered at the same time-point.
- **Pooled data**: A combination of time series data and cross-sectional data.

"Time series analysis is a statistical technique dealing in time series data, or trend analysis."

- Consider an example of a restaurant in which prediction is made on the number of customers as when will more customers appear in the restaurant at a specified time duration based on the previous appearance of customers with time.

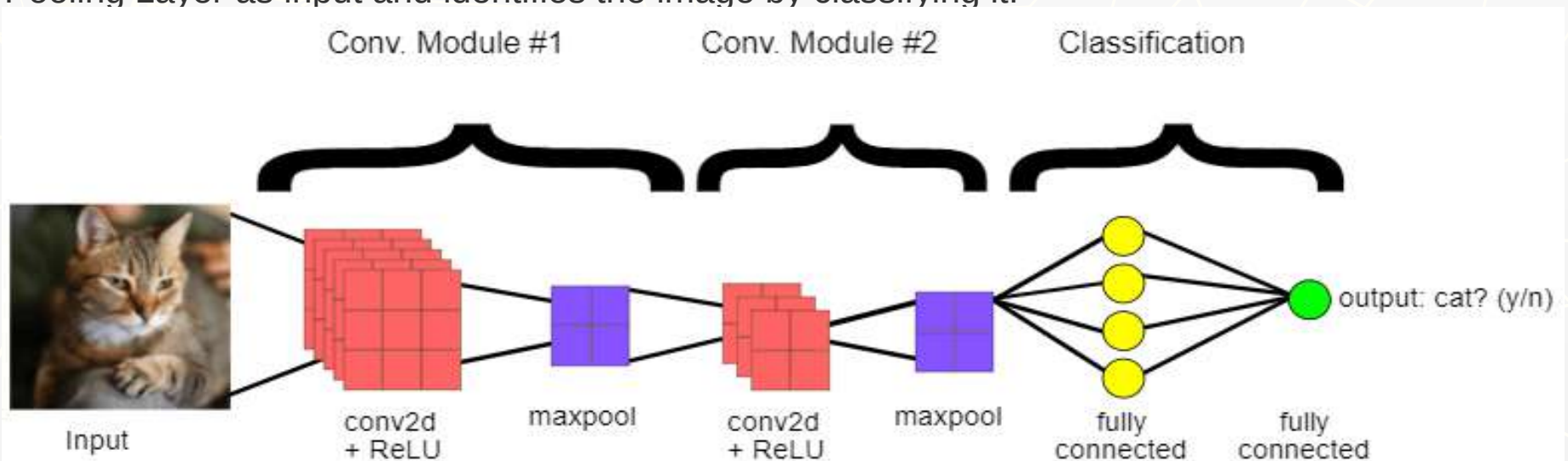
Deep Learning Algorithms

The Deep Learning Algorithms are as follows:

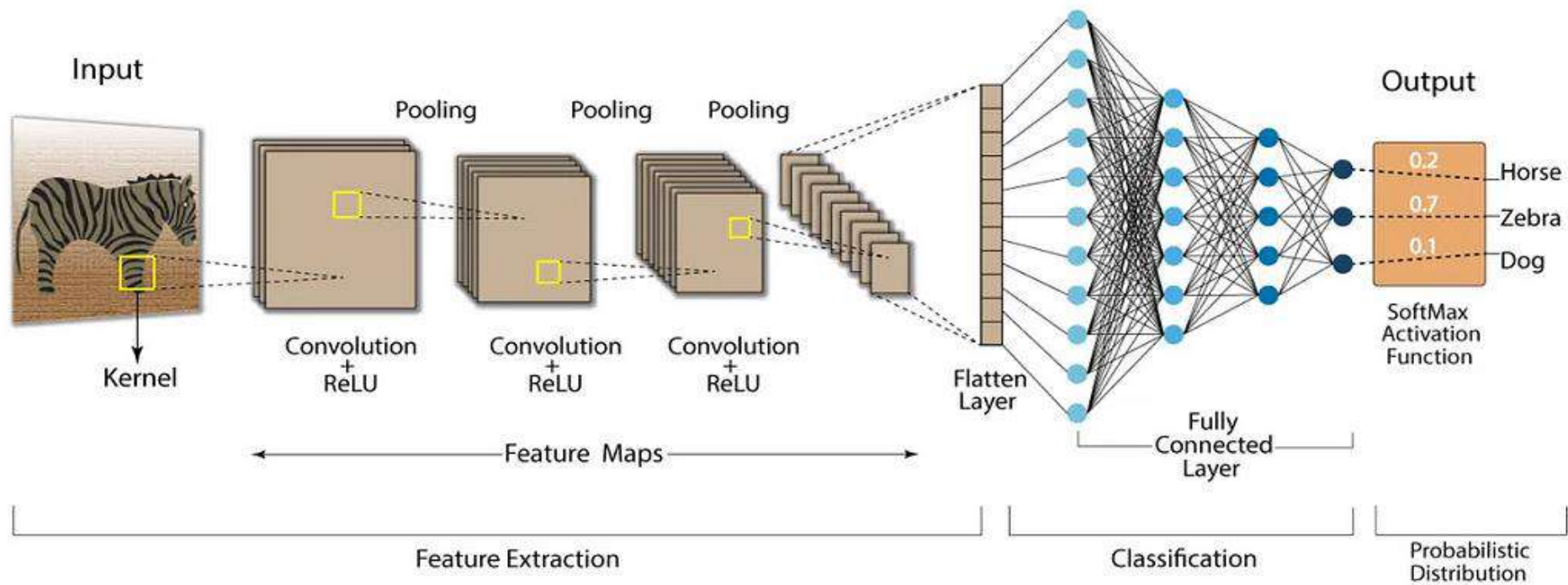
1. Convolutional Neural Networks (CNNs)

CNN's popularly known as **ConvNets** majorly consists of several layers and are specifically used for image processing and detection of objects. It was developed in **1998** by **Yann LeCun** and was first called **LeNet**. Back then, it was developed to recognize digits and zip code characters. CNNs have wide usage in identifying the image of the satellites, medical image processing, series forecasting, and anomaly detection.

CNNs process the data by passing it through multiple layers and extracting features to exhibit convolutional operations. The **Convolutional Layer** consists of **Rectified Linear Unit (ReLU)** that outlasts to rectify the feature map. The **Pooling layer** is used to rectify these feature maps into the next feed. Pooling is generally a sampling algorithm that is down-sampled and it reduces the dimensions of the feature map. Later, the result generated consists of **2-D arrays** consisting of **single, long, continuous, and linear vector** flattened in the map. The next layer i.e., called **Fully Connected Layer** which forms the flattened **matrix** or **2-D array** fetched from the Pooling Layer as input and identifies the image by classifying it.



Convolution Neural Network (CNN)



1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

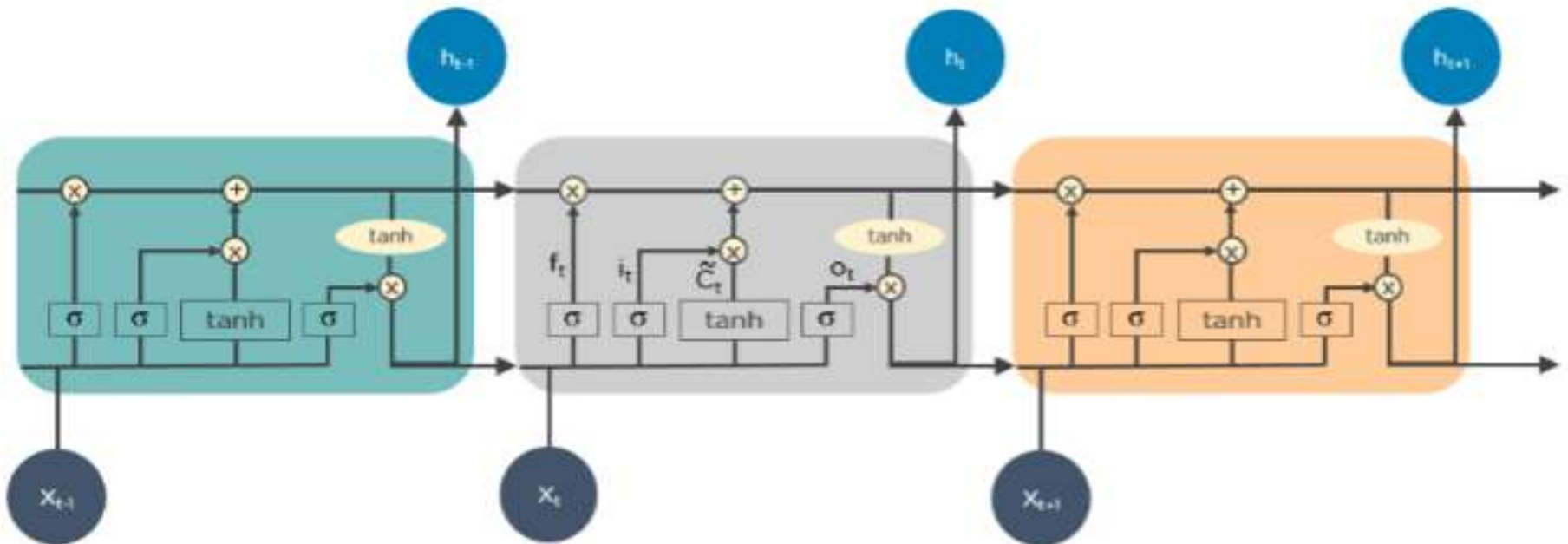
4		

Convolved Feature

Deep Learning Algorithms

2. Long Short Term Memory Networks (LSTMs)

LSTMs can be defined as **Recurrent Neural Networks** (RNN) that are programmed to learn and adapt for dependencies for the long term. It can memorize and recall past data for a greater period and by default, it is its sole behavior. LSTMs are designed to retain over time and henceforth they are majorly used in time series predictions because they can restrain memory or previous inputs. This analogy comes from their **chain-like** structure consisting of **four** interacting layers that communicate with each other differently. Besides applications of time series prediction, they can be used to construct **speech recognizers**, **development in pharmaceuticals**, and composition of **music loops** as well. LSTM work in a sequence of events. First, they don't tend to remember irrelevant details attained in the previous state. Next, they update certain cell-state values selectively and finally generate certain parts of the cell-state as output. Below is the diagram of their operation.

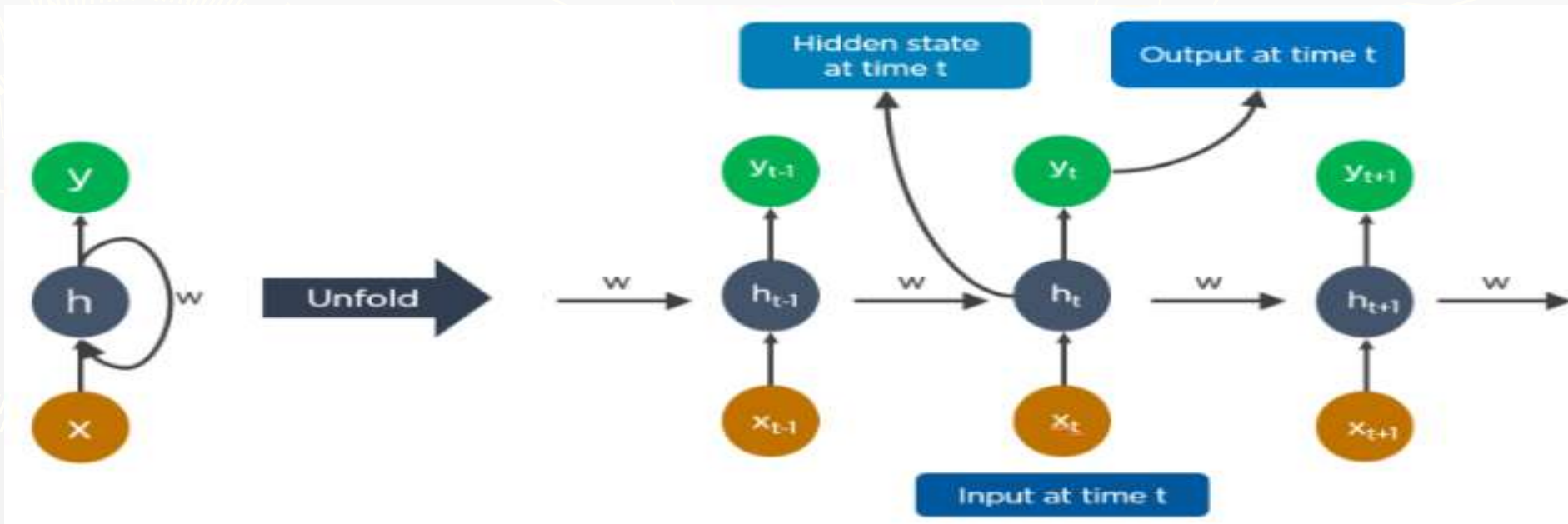


Deep Learning Algorithms

3. Recurrent Neural Networks (RNNs)

Recurrent Neural Networks or RNNs consist of some directed connections that form a cycle that allow the input provided from the LSTMs to be used as input in the current phase of RNNs. These inputs are deeply embedded as inputs and enforce the memorization ability of LSTMs lets these inputs get absorbed for a period in the internal memory. RNNs are therefore dependent on the inputs that are preserved by LSTMs and work under the synchronization phenomenon of LSTMs. RNNs are mostly used in captioning the image, time series analysis, recognizing handwritten data, and translating data to machines.

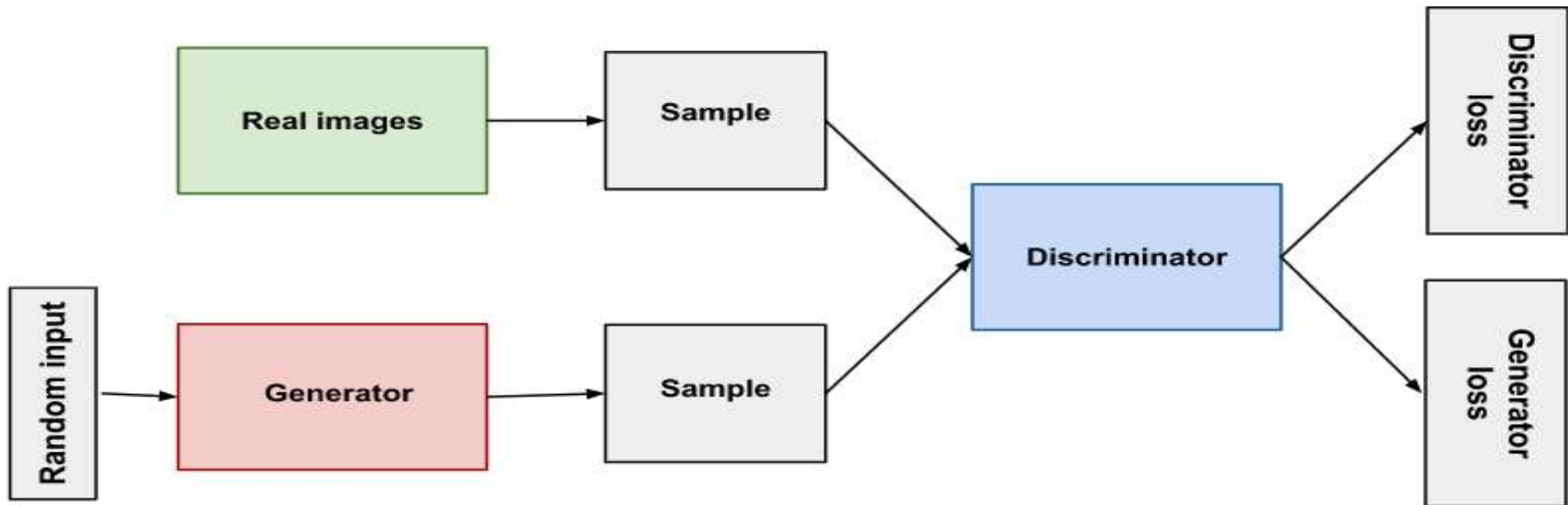
RNNs follow the work approach by putting output feeds **($t-1$)** time if the time is defined as **t** . Next, the output determined by t is feed at input time **$t+1$** . Similarly, these processes are repeated for all the input consisting of any length. There's also a fact about RNNs is that they store historical information and there's no increase in the input size even if the model size is increased. RNNs look something like this when unfolded.



Deep Learning Algorithms

4. Generative Adversarial Networks (GANs)

GANs are defined as deep learning algorithms that are used to generate new instances of data that match the training data. GAN usually consists of two components namely a **generator** that learns to generate false data and a **discriminator** that adapts itself by learning from this false data. Over some time, GANs have gained immense usage since they are frequently being used to clarify **astronomical images** and simulate **lensing** the gravitational dark matter. It is also used in **video games** to increase graphics for **2D** textures by recreating them in higher resolution like **4K**. They are also used in creating **realistic cartoons character** and also rendering human faces and **3D object rendering**. GANs work in simulation by generating and understanding the fake data and the real data. During the training to understand these data, the generator produces different kinds of fake data where the discriminator quickly learns to adapt and respond to it as false data. GANs then send these recognized results for updating. Consider the below image to visualize the functioning.

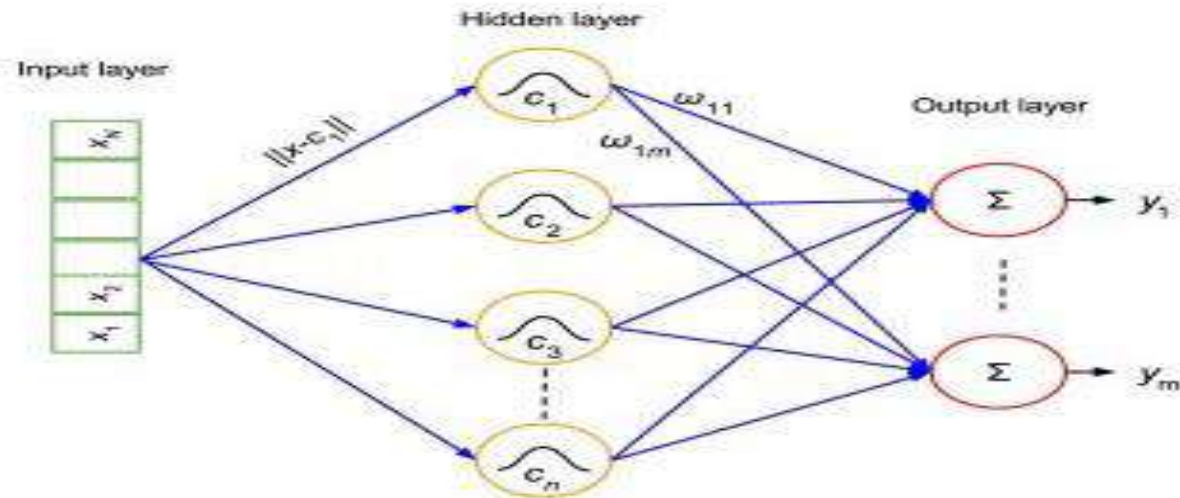


Deep Learning Algorithms

5. Radial Basis Function Networks (RBFNs)

RBFNs are specific types of neural networks that follow a feed-forward approach and make use of radial functions as activation functions. They consist of **three** layers namely the **input layer**, **hidden layer**, and **output layer** which are mostly used for **time-series prediction**, **regression testing**, and **classification**.

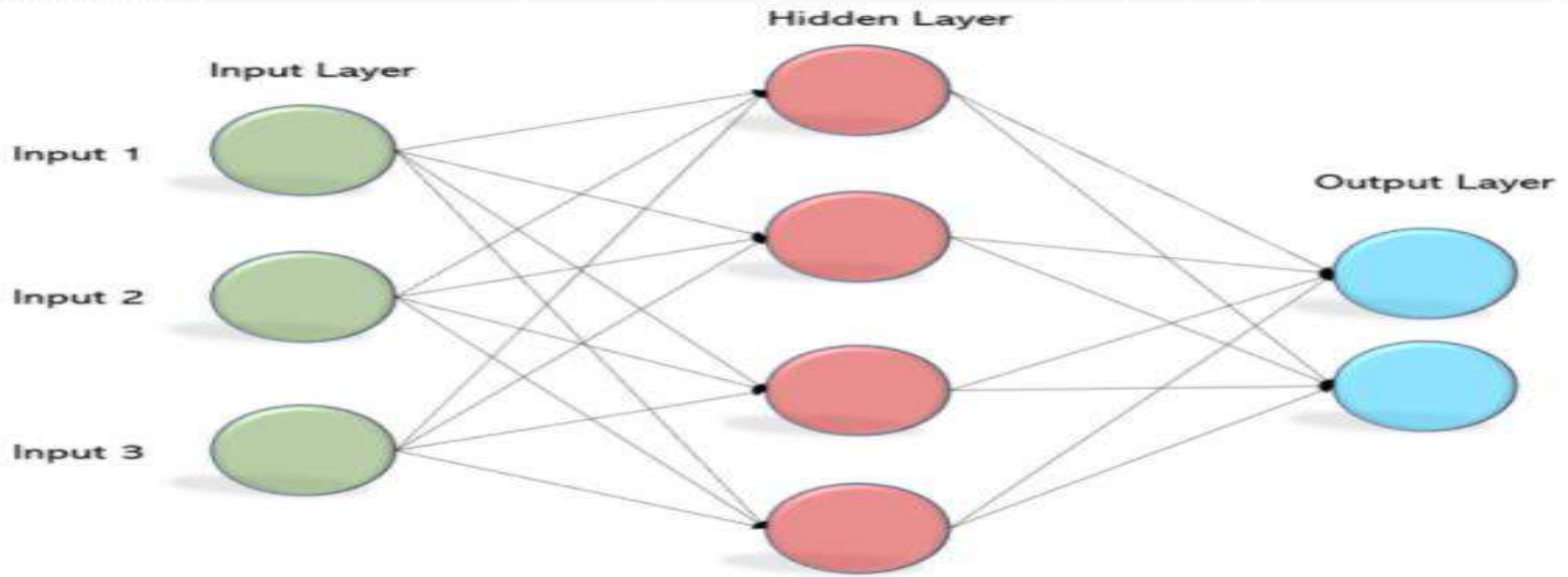
RBFNs do these tasks by measuring the similarities present in the training data set. They usually have an input vector that feeds these data into the input layer thereby confirming the identification and rolling out results by comparing previous data sets. Precisely, the input layer has **neurons** that are sensitive to these data and the nodes in the layer are efficient in classifying the class of data. Neurons are originally present in the hidden layer though they work in close integration with the input layer. The hidden layer contains **Gaussian transfer** functions that are inversely proportional to the distance of the output from the neuron's center. The output layer has linear combinations of the **radial-based** data where the Gaussian functions are passed in the neuron as parameter and output is generated. Consider the given image below to understand the process thoroughly.



Deep Learning Algorithms

6. Multilayer Perceptrons (MLPs)

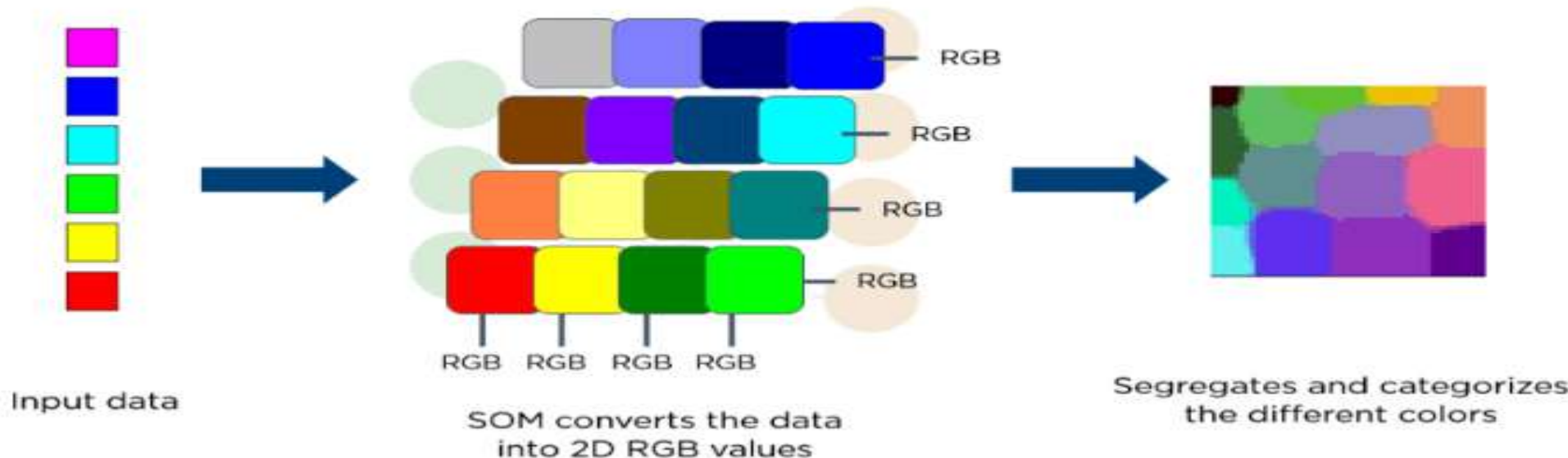
MLPs are the base of deep learning technology. It belongs to a class of feed-forward neural networks having various layers of **perceptrons**. These perceptrons have various activation functions in them. MLPs also have connected input and output layers and their number is the same. Also, there's a layer that remains hidden amidst these two layers. MLPs are mostly used to build **image and speech recognition** systems or some other types of the **translation software**. The working of MLPs starts by feeding the data in the input layer. The neurons present in the layer form a graph to establish a connection that passes in one direction. The weight of this input data is found to exist between the hidden layer and the input layer. MLPs use activation functions to determine which nodes are ready to fire. These activation functions include **tanh** function, **sigmoid** and **ReLU**s. MLPs are mainly used to train the models to understand what kind of co-relation the layers are serving to achieve the desired output from the given data set. See the below image to understand better.



Deep Learning Algorithms

7. Self Organizing Maps (SOMs)

SOMs were invented by **Teuvo Kohonen** for achieving data visualization to understand the dimensions of data through artificial and self-organizing neural networks. The attempts to achieve data visualization to solve problems are mainly done by what humans cannot visualize. These data are generally high-dimensional so there are lesser chances of human involvement and of course less error. SOMs help in visualizing the data by initializing weights of different nodes and then choose random vectors from the given training data. They examine each node to find the relative weights so that dependencies can be understood. The winning node is decided and that is called **Best Matching Unit (BMU)**. Later, SOMs discover these winning nodes but the nodes reduce over time from the sample vector. So, the closer the node to BMU more is the more chance to recognize the weight and carry out further activities. There are also multiple iterations done to ensure that no node closer to BMU is missed. One example of such is the **RGB color combinations** that we use in our daily tasks. Consider the below image to understand how they function.

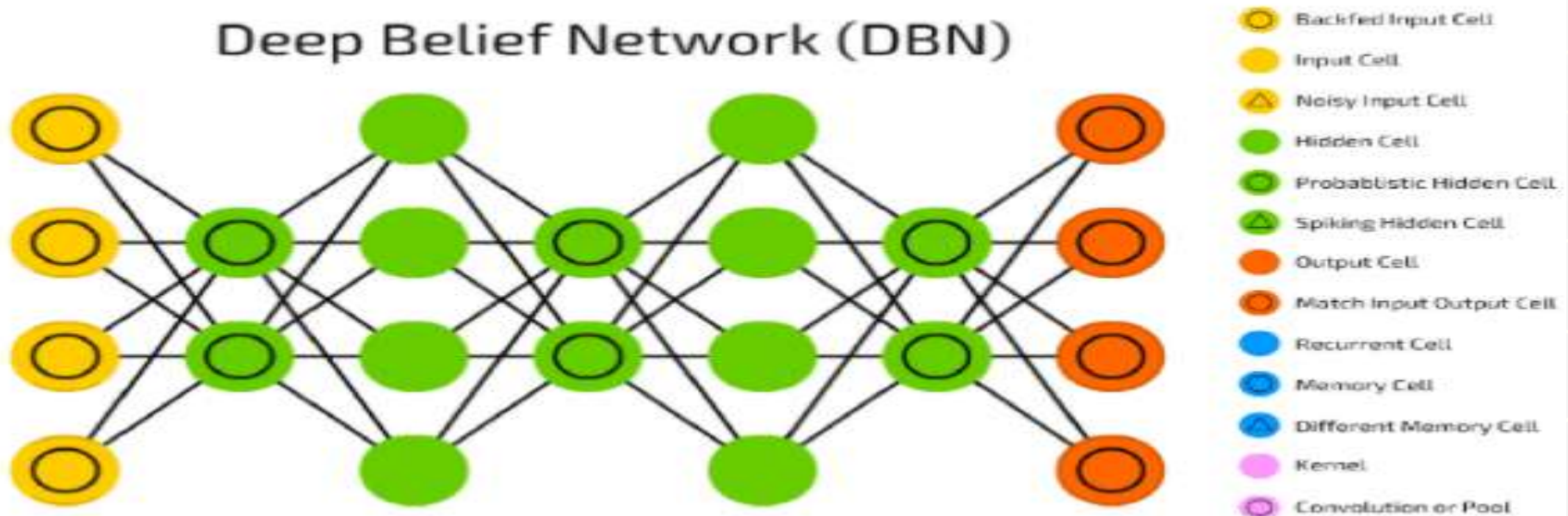


Deep Learning Algorithms

8. Deep Belief Networks (DBNs)

DBNs are called generative models because they have various layers of latent as well as stochastic variables. The latent variable is called a **hidden unit** because they have binary values. DBNs are also called **Boltzmann Machines** because the **RBM** layers are stacked over each other to establish communication with previous and consecutive layers. DBNs are used in applications like video and image recognition as well as capturing motional objects. DBNs are powered by **Greedy algorithms**. The layer to layer approach by leaning through a **top-down** approach to generate weights is the most common way DBNs function. DBNs use step by step approach of **Gibbs** sampling on the hidden **two-layer** at the top. Then, these stages draw a sample from the visible units using a model that follows the ancestral sampling method. DBNs learn from the values present in the latent value from every layer following the **bottom-up** pass approach.

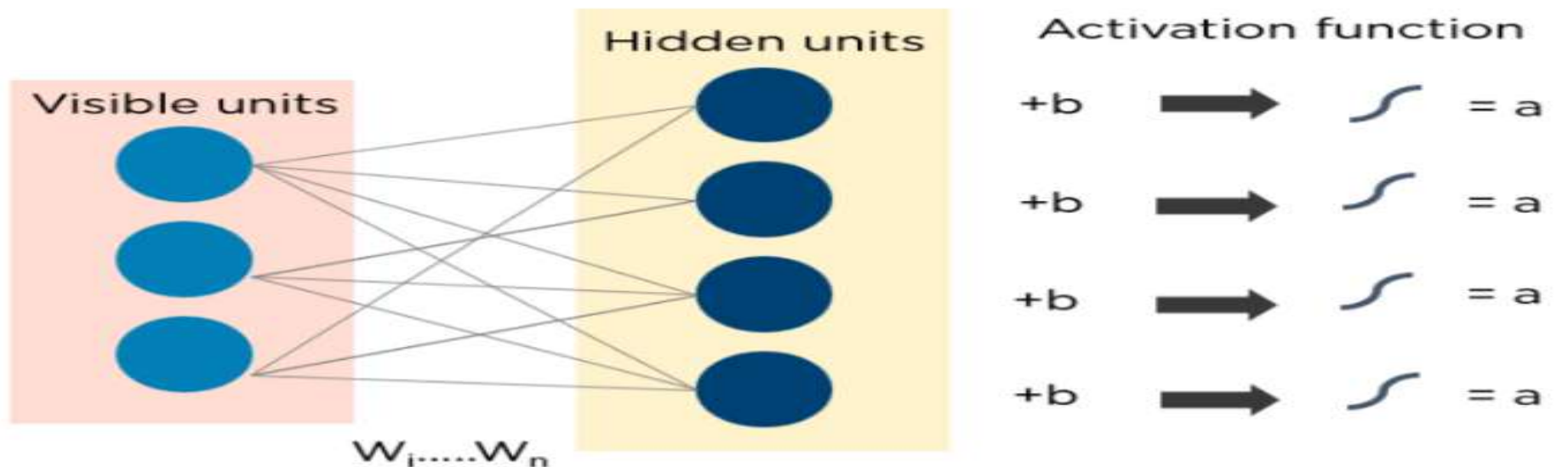
Deep Belief Network (DBN)



Deep Learning Algorithms

9. Restricted Boltzmann Machines (RBMs)

RBMs were developed by **Geoffrey Hinton** and resemble stochastic neural networks that learn from the probability distribution in the given input set. This algorithm is mainly used in the field of dimension **reduction**, **regression** and **classification**, **topic modeling** and are considered the building blocks of DBNs. RBIs consist of two layers namely the **visible layer** and the **hidden layer**. Both of these layers are connected through hidden units and have bias units connected to nodes that generate the output. Usually, RBMs have two phases namely **forward pass** and **backward pass**. The functioning of RBMs is carried out by accepting inputs and translating them to numbers so that inputs are encoded in the forward pass. RBMs take into account the weight of every input, and the backward pass takes these input weights and translates them further into reconstructed inputs. Later, both of these translated inputs, along with individual weights, are combined. These inputs are then pushed to the visible layer where the activation is carried out, and output is generated that can be easily reconstructed. To understand this process, consider the below image.

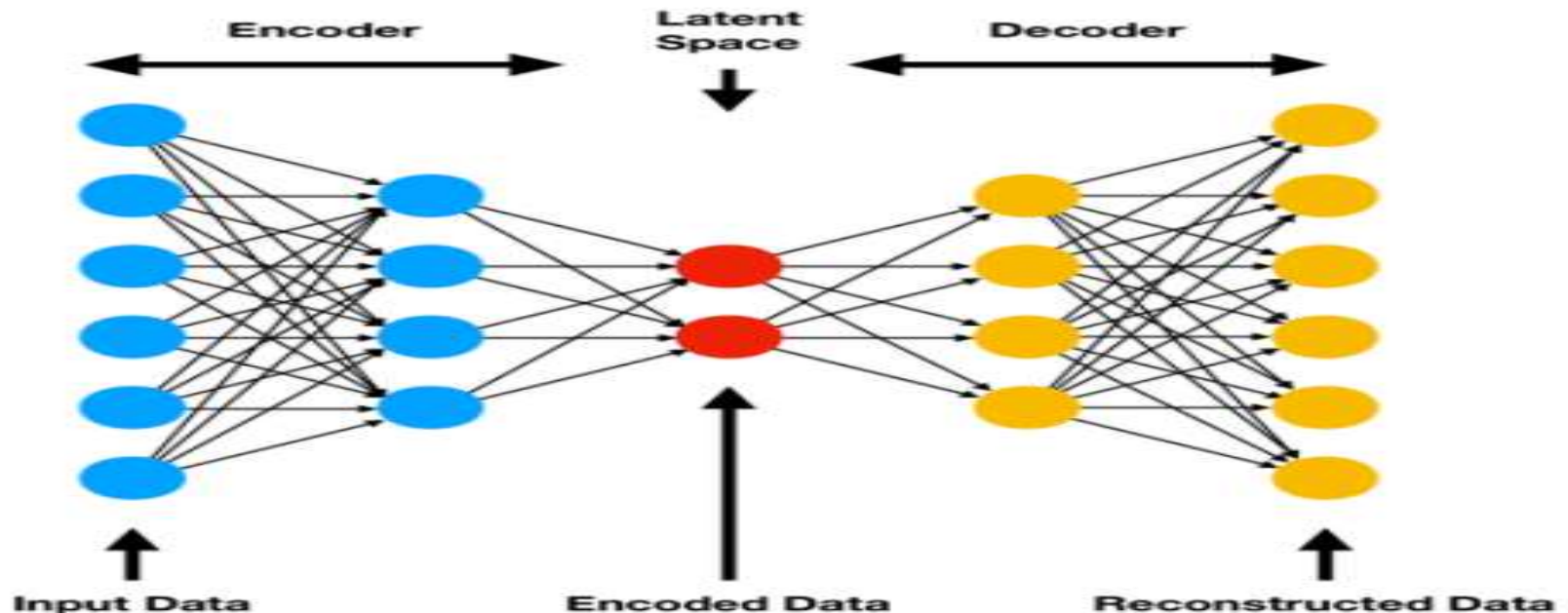


Deep Learning Algorithms

Auto encoders

Auto encoders are a special type of neural network where inputs are outputs are found usually identical. It was designed to primarily solve the problems related to unsupervised learning. Auto encoders are highly trained neural networks that **replicate** the data. It is the reason why the input and output are generally the same. They are used to achieve tasks like **pharma discovery, image processing, and population prediction**.

Auto encoders constitute three components namely the **encoder**, the **code**, and the **decoder**. Auto encoders are built in such a structure that they can receive inputs and transform them into various representations. The attempts to copy the original input by reconstructing them is more accurate. They do this by encoding the image or input, reduce the size. If the image is not visible properly they are passed to the neural network for clarification. Then, the clarified image is termed a reconstructed image and this resembles as accurate as of the previous image. To understand this complex process, see the below-provided image.



Types of Convolutional Neural Networks (CNNs)

There are a class of deep learning models primarily used for image processing, computer vision, and other tasks involving grid-like data. There are several types of CNNs, each designed for specific tasks or improvements over traditional architectures. Here are some common types:

1. Vanilla CNN (Traditional CNN)

The basic CNN architecture consisting of:

- Convolutional layers (for feature extraction)

- Pooling layers (for downsampling)

- Fully connected layers (for classification)

Example: LeNet-5 (one of the earliest CNNs for digit recognition).

2. AlexNet

A deeper and more advanced version of the traditional CNN.
Introduced ReLU activation, dropout, and data augmentation.
Won the ImageNet competition in 2012.

3. VGGNet (Visual Geometry Group)

Known for its simplicity and depth.
Uses small 3x3 convolutional filters stacked in layers.
Popular variants: VGG-16 and VGG-19.

4. GoogLeNet (Inception Network)

Introduced the "Inception module," which uses multiple filter sizes (1x1, 3x3, 5x5) in parallel.
Reduces computational cost and improves feature extraction.
Example: InceptionV1, InceptionV3.

Types of Convolutional Neural Networks (CNNs)

5. ResNet (Residual Network)

- Introduced residual connections (skip connections) to address the vanishing gradient problem in very deep networks.
- Enabled training of networks with hundreds of layers.
- Example: ResNet-50, ResNet-101.

6. DenseNet (Densely Connected Network)

- Each layer is connected to every other layer in a feed-forward manner.
- Improves feature reuse and reduces the number of parameters.

7. MobileNet

- Designed for mobile and embedded devices.
- Uses depthwise separable convolutions to reduce computational cost.
- Lightweight and efficient.

8. EfficientNet

- Uses a compound scaling method to balance model depth, width, and resolution.
- Achieves state-of-the-art performance with fewer parameters.

9. U-Net

- Designed for image segmentation tasks.
- Uses an encoder-decoder architecture with skip connections.
- Commonly used in medical image analysis.

10. YOLO (You Only Look Once)

- A CNN-based architecture for real-time object detection. Processes the entire image in a single forward pass.

Types of Convolutional Neural Networks (CNNs)

11. R-CNN (Region-Based CNN)

- Used for object detection.
- Variants: Fast R-CNN, Faster R-CNN, Mask R-CNN.
- Generates region proposals and classifies them.

12. Capsule Networks (CapsNet)

- An alternative to traditional CNNs.
- Uses "capsules" to capture spatial relationships between features.
- Addresses limitations of CNNs in handling pose and orientation.

13. Siamese Networks

- Used for tasks like image similarity and face recognition.
- Consists of two identical CNNs sharing weights.

14. GANs (Generative Adversarial Networks)

- Consists of two CNNs: a generator and a discriminator.
- Used for generating realistic images, data augmentation, etc.

15. Transformer-based CNNs

- Combines CNNs with transformer architectures (e.g., Vision Transformers, ViT).
- Achieves state-of-the-art results in image classification.

16. SqueezeNet

- Designed to have fewer parameters while maintaining accuracy.
- Uses 1x1 convolutions and fire modules.

17. Xception

An extension of Inception networks.

Uses depthwise separable convolutions extensively.

What is new in deep learning?

18. ShuffleNet

Designed for mobile devices with limited computational power.
Uses channel shuffling to improve efficiency.

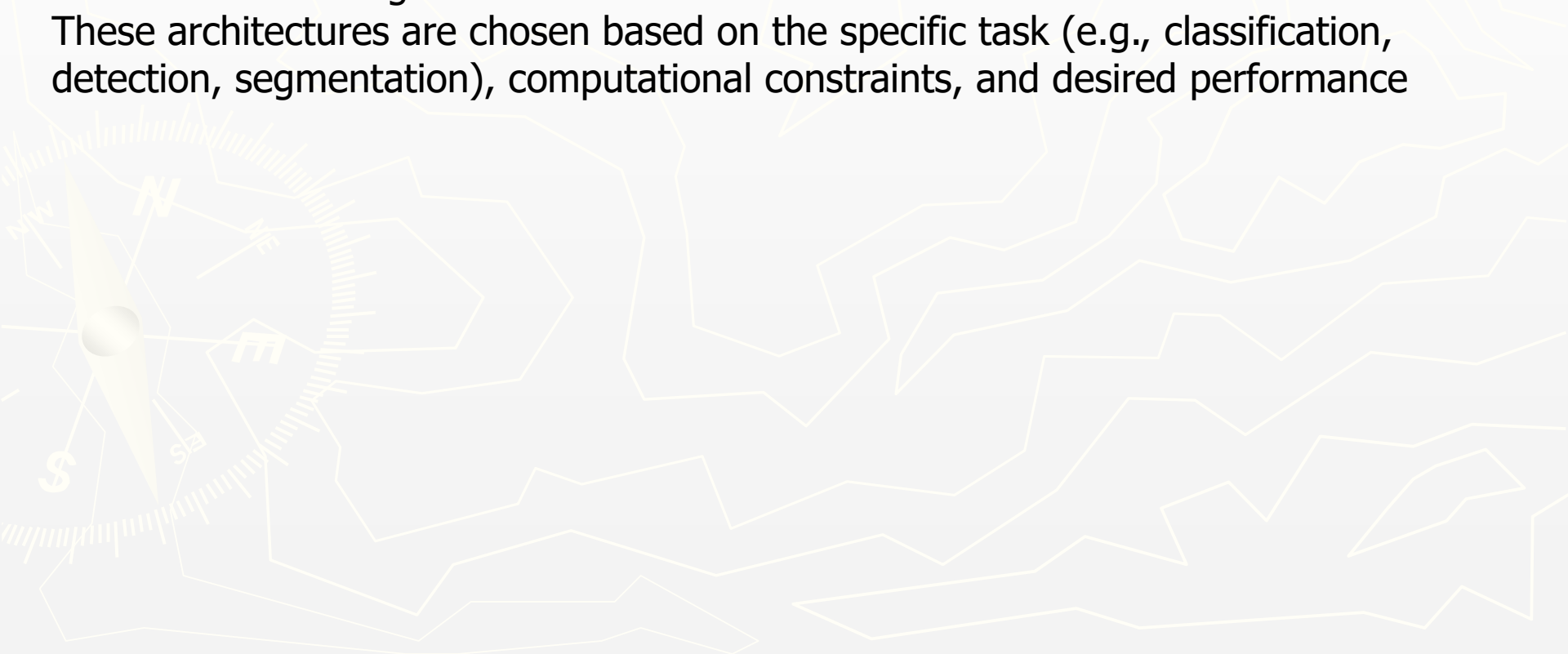
19. NASNet (Neural Architecture Search Network)

Automatically designed using neural architecture search (NAS).
Optimized for specific tasks.

20. Swin Transformer

A hierarchical vision transformer that uses shifted windows for efficient computation.
Combines the strengths of CNNs and transformers.

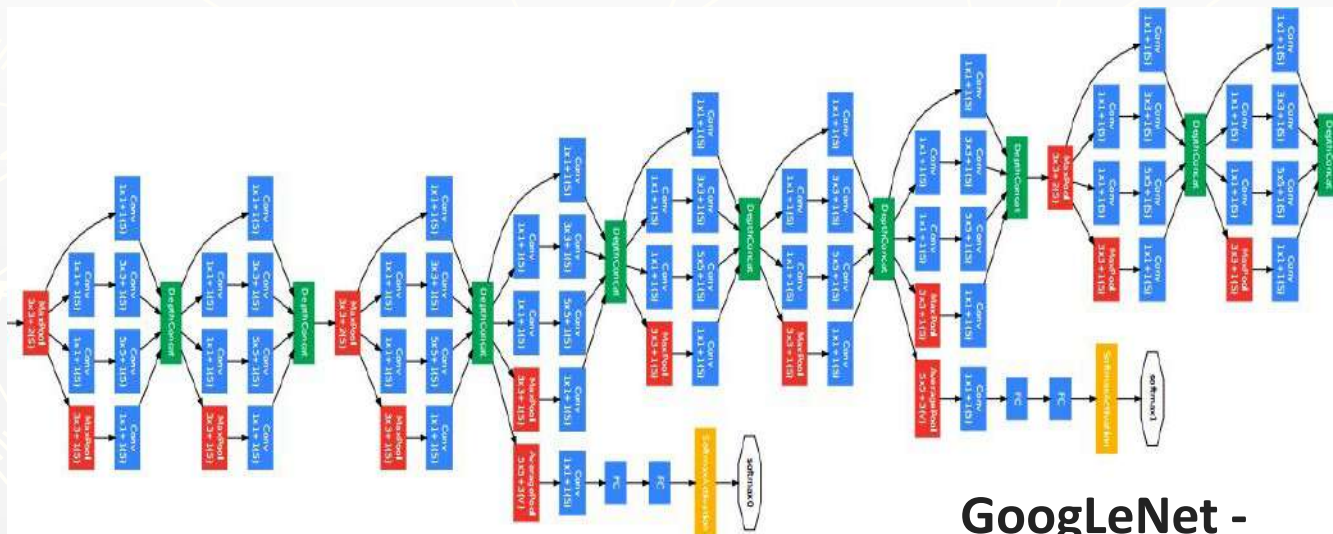
These architectures are chosen based on the specific task (e.g., classification, detection, segmentation), computational constraints, and desired performance



What is new in deep learning?

What is new (among other things) is a learning algorithm called **backpropagation** which allows to train **deep neural nets**.

State-of-the-art networks can have **over 200 layers!**



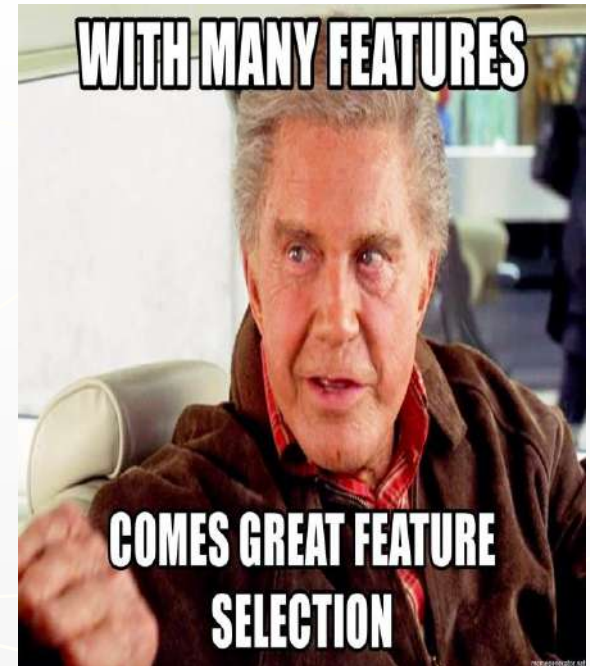
GoogLeNet -
2014

Difference classical ML vs. Deep Learning

Classical ML methods don't handle high dimensionality well.

⇒ **dimensionality reduction** & **feature selection**

Deep neural nets learn **compact representations** of data even in a high dimensionality/sparse setting - **no feature engineering** required!



Unsupervised Learning: Generative Adversarial Nets (GAN)



2014



2015



2016



2017



2018

NVIDIA

None of these images were taken in the real world!

Unsupervised Learning: Generative Adversarial Nets (GAN)



DeepMind - 2018

Which of these images was generated?

Unsupervised Learning: Language Generation

OpenAI, February 14 2019: Better Language Models and Their Implications

SYSTEM PROMPT
(HUMAN-WRITTEN)

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

MODEL COMPLETION
(MACHINE-WRITTEN,
10 TRIES)

The scientist named the population, after their distinctive horn, Ovid's Unicorn. These four-horned, silver-white unicorns were previously unknown to science.

Now, after almost two centuries, the mystery of what sparked this odd phenomenon is finally solved.

Dr. Jorge Pérez, an evolutionary biologist from the University of La Paz, and several companions, were exploring the Andes Mountains when they found a small valley, with no other animals or humans. Pérez noticed that the valley had what appeared to be a natural fountain, surrounded by two peaks of rock and silver snow.

Pérez and the others then ventured further into the valley. "By

So why not use Deep Learning for everything?



- ▶ There are reasons why we don't only use DL:
- Necessary data no available
- Computational power not available
- Harder to interpret results
- Deep networks can be fooled:

